

OUTPUTS

```
--- Decision Tree Comparison (Iris) ---  
Custom Decision Tree Accuracy: 1.0000  
Scikit-learn Decision Tree Accuracy: 1.0000
```

```
--- Ensemble Methods (Wine Classification) ---  
Decision Tree F1 Score: 0.9425  
Random Forest F1 Score: 1.0000
```

```
--- Hyperparameter Tuning (Random Forest Classifier) ---  
Best Parameters: {'max_depth': None, 'min_samples_split': 2, 'n_estimators': 100}  
Best F1 Score: 0.9862945382658644
```

```
--- Regression Models (Wine) ---  
Decision Tree MSE: 0.1667  
Random Forest MSE: 0.0648
```

```
--- Hyperparameter Tuning (Random Forest Regressor) ---  
Best Parameters: {'n_estimators': 300, 'min_samples_split': 2, 'max_depth': 10}  
Best Score (Negative MSE): -0.05090262509850276
```